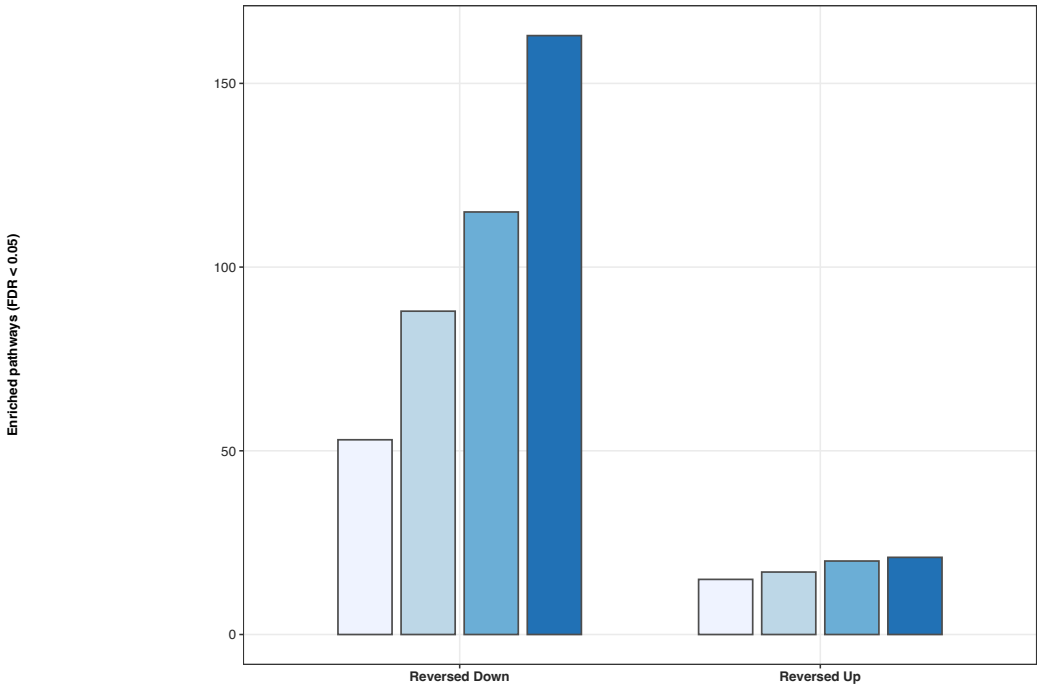
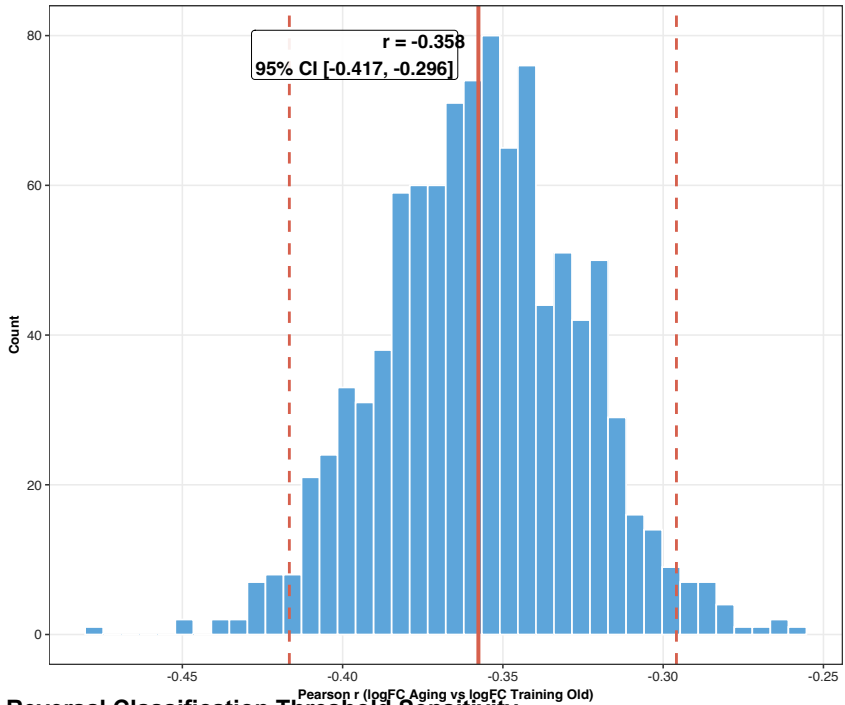


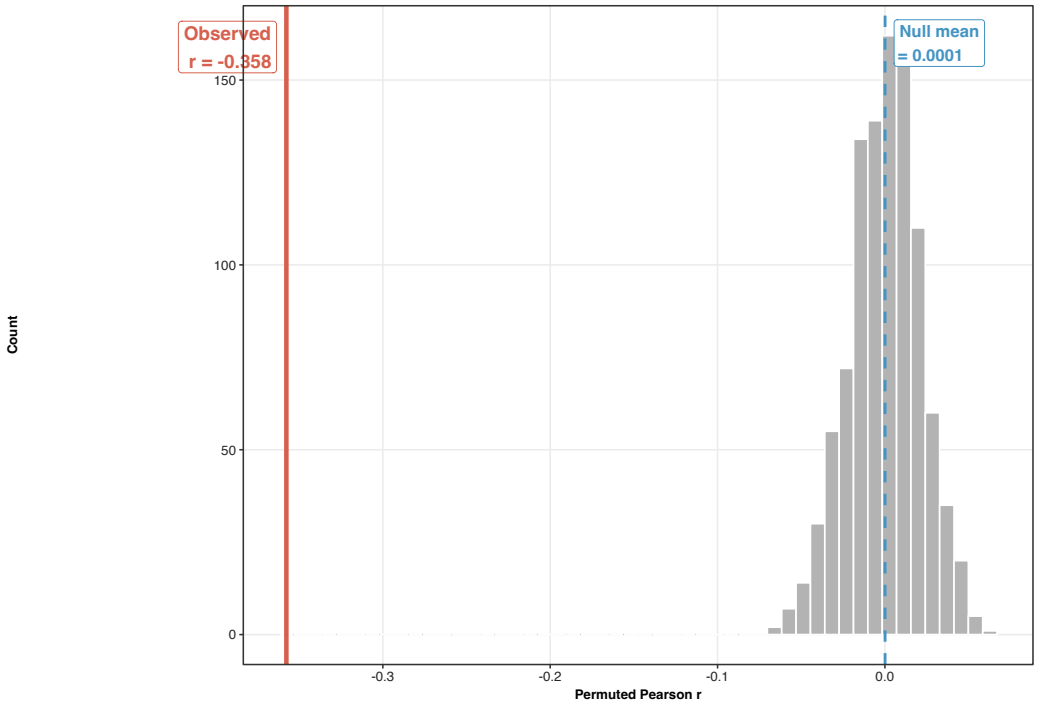
**A** **ORA Dedup Sensitivity (Reversal Quadrants)**  
*Hypergeometric ORA | 2103 universe | Reversed quadrants only*



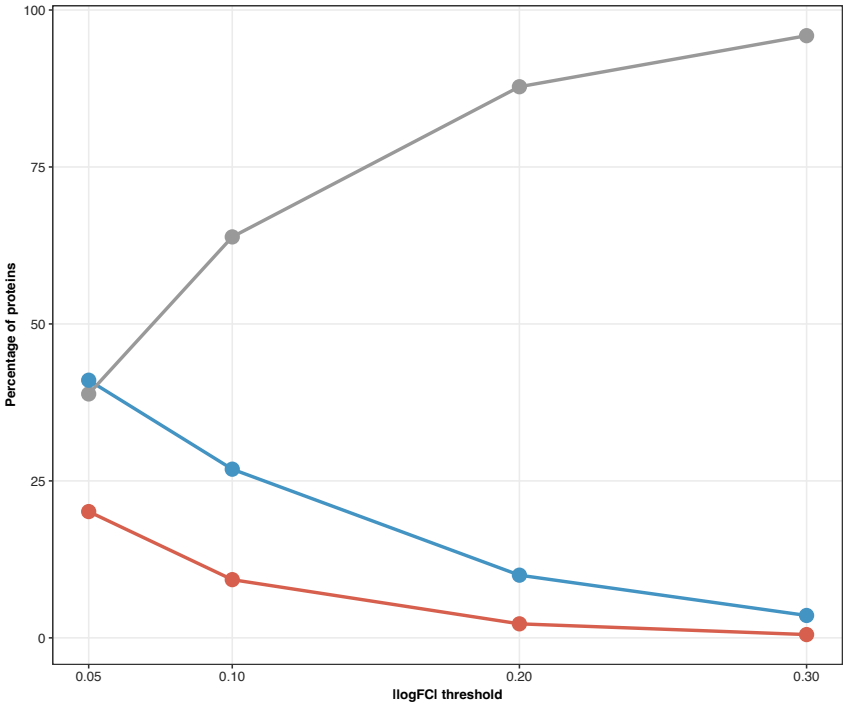
**B** **Reversal Pearson r Bootstrap**  
*boot::boot() R = 1000 | n = 2103 | r = -0.358 [-0.417, -0.296]*



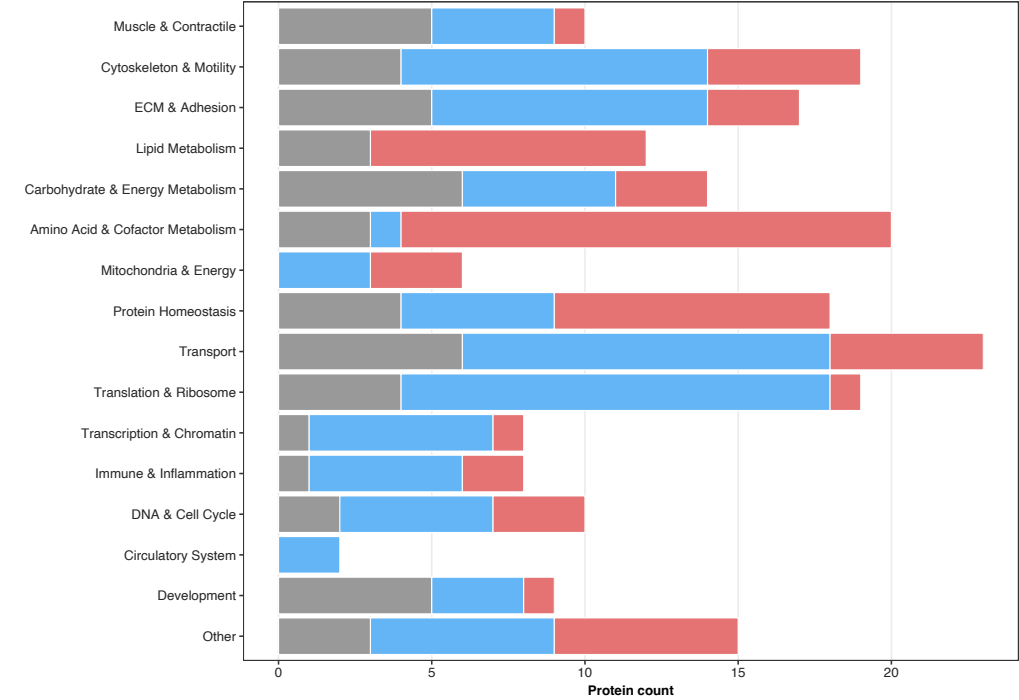
**C** **Circularity Diagnostic: Protein-Permuted Null**  
*Observed r = -0.358 | Null mean = 0.0001 | p\_perm = 0.000*



**D** **Reversal Classification Threshold Sensitivity**  
*logFC thresholds: 0.05, 0.1, 0.2, 0.3 | stable reversal %*



**E** **GO Slim Category Distribution (Reversal)**  
*DEPs (π < 0.05) | 210 proteins | 16 categories*



**F** **fry Leading-Edge Proteins (Reversal Drivers)**  
*Top 25 by |t(Training Old)| | colored by aging direction*

